
Approximate Inference for Time-varying Interactions and Macroscopic Dynamics of Neural Populations

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Abstract

The maximum entropy principle has been successfully employed to explain stationary spiking activity of a neural population by using fewer parameters, such as spike-rates of individual neurons and low-order interactions among neurons, than the number of possible activity patterns. Modeling activity of cortical circuitries *in vivo*, however, has been challenging because both spike-rates and interactions can change according to sensory stimulation, behavior, or an internal state of the brain. However, previous approaches modeling the dynamics of neural interactions suffer from computational cost; therefore, its application was limited to only a dozen neurons. Here by introducing multiple analytic approximation methods to a state-space model of neural population activity, we make it possible to estimate dynamic pairwise interactions of up to 60 neurons. More specifically, we applied the pseudolikelihood approximation to the state-space model, and combined it with the TAP mean-field or Bethe approximation to make the sequential Bayesian estimation of the model parameters possible. The large-scale analysis allows us to investigate dynamics of macroscopic properties of neural circuitries underlying stimulus processing and behavior. We show that the model accurately estimates dynamics of network properties such as sparseness, entropy, and heat capacity by simulated data, and demonstrate utilities of these measures by analyzing activity of monkey V4 neurons.

Introduction

Activity patterns of neuronal populations are constrained by biological mechanisms such as biophysical properties of each neuron (e.g., synaptic integration and spike generation [1, 2]) and their anatomical connections [3]. The characteristic correlations imposed by the biological mechanisms interplay with statistics of sensory inputs, and influence how the sensory information is represented in the population activity [4–6]. Thus accurate assessment of the neural correlations in ongoing

and evoked activities is a key to understand the underlying biological mechanisms and their coding principles.

The number of possible activity patterns increases combinatorially with the number of neurons analyzed. The maximum entropy (ME) principle and derived ME models - such as the pairwise ME model or the Ising model - have been successfully used to explain neural population activities using fewer activity features such as event rates or correlations among subset neurons [7, 8]. For a large system, however, exact inference of these models becomes computationally infeasible. Thus researchers have employed approximation methods [9–15]. Another fundamental problem for the conventional ME models is the stationary assumption: The models assume temporarily constant features in neural activity patterns such as spike rates of individual neurons. This assumption is not valid, e.g., when *in vivo* activity is recorded while an animal performs a behavioral task. Ignoring such dynamics might result in erroneous model estimates and misleading interpretations [16–19]. Moreover neural correlations likely organize dynamically during behavior and cognition, which can be independent from changes in the spike rates of individual neurons [20]. The time-dependence of neural activity may be explained by including stimulus signals in the model, e.g., for analyses of early sensory cells [21]. However, the approach may become impractical when analyzing neurons in higher brain areas in which receptive fields of neurons are not easily characterized.

Recently several researchers proposed methods for simultaneously estimating dynamic neural interactions without modeling receptive fields of neurons [22–27]. For example, the approaches proposed in [22, 23, 26, 28] suggested estimation of the time-varying neural interactions using a general smoothness prior. In addition to the point estimates of interaction parameters, the state-space model [22, 26, 28] provides credible intervals of those estimates through the recursive Bayesian fitting algorithm. Nevertheless, as previously mentioned, these methods are restricted by their computational cost. Therefore, they have been utilized to analyze only small populations ($N \leq 15$). Recent advances in electrophysiological and optical recording techniques from a large number of neurons *in vivo* under free moving or virtual reality settings challenge these analysis methods. Alternatively, unsupervised methods that aim to capture low-dimensional, time-dependent latent structure of the simultaneous neural activity have been proposed [29–31] (See also [32–34] for other dimension reduction methods for neuroscience data). However, methods that directly estimate time-varying interactions of a large neural population are also needed because they serve as a benchmark against the dimension reduction methods, and may be necessary to accurately assess macroscopic properties of the neurons.

Here by combining the state-space model proposed in [22, 26, 28] with analytic approximation methods, we provide a framework for estimating interactions of neuronal populations consisting of up to 60 neurons. The Bayesian analysis methods for larger networks of neurons allow us to better understand macroscopic states of a neural population, such as entropy, free energy and sensitivity, all in a time-resolved manner and with confidence intervals. Thus the model provides a new way to investigate effects of stimuli and behavior on activity of neuronal populations. It is expected to provide observations that give us insights into the underlying circuitry and its computation.

Materials and Methods

To clarify the problem of large-scale analysis on dynamic population activity, we first formulate the state-space model and its estimation method originally investigated in [22, 26] in the next subsection. Then we describe how to introduce approximation methods to the state-space model in order to overcome the limitation of the model and make the large-scale analysis possible. The custom-made Python programs is provided on GitHub (https://github.com/christiando/ssl_lib).

The state-space analysis of neural population activity

Spike data To investigate how neuronal activities realize perception, cognition, and behavior, neurophysiologists record timing of neuronal spiking activity over the course of a behavioral paradigm designed to test specific hypotheses. Typically, these experiments are repeated multiple times under the same experimental conditions to uncover common neuronal dynamics related to the behavioral paradigm from stochastic spiking activities. We assume that neural data is composed of repeated measurements (R times) of spike timing recorded from N neurons simultaneously. Hereafter rep-

etition is termed trial. To analyze activity patterns of the neurons, we discretize the parallel spike sequences into T time bins with bin size Δ , and represent the population activity by a set of binary variables. Namely, for neurons $n = 1, \dots, N$, time bins $t = 1, \dots, T$, and trials $r = 1, \dots, R$, we represent the neural activity by a binary variable $X_n^{r,t}$, where $X_n^{r,t} = 1$ when neuron n spiked in time bin t and trial r ; and $X_n^{r,t} = 0$ otherwise. The whole data is represented by a $N \times R \times T$ dimensional binary matrix. The activity pattern of N neurons at time bin t and trial r is represented by a vector, $\mathbf{X}^{r,t} = (X_1^{r,t}, \dots, X_N^{r,t})'$. Similarly, $\mathbf{X}^t = (\mathbf{X}^{1,t}, \dots, \mathbf{X}^{R,t})$ summarizes observations for all neurons $1, \dots, N$ and all trials $1, \dots, R$ at time bin t . Finally, $\mathbf{X}^{t_1:t_2} = (\mathbf{X}^{t_1}, \dots, \mathbf{X}^{t_2})$ denotes the observations from time bin t_1 to t_2 .

State-space model of neural population activity The state-space model of dynamic population activity is composed of an observation model and a state model. The observation model specifies the probability distribution of population activity patterns using state variables, whereas the latter dictates how those state variables change. Here we construct the observation model using the exponential family distribution considering up to pairwise interactions of neurons' activities,

$$p(\mathbf{x}|\boldsymbol{\theta}_t) = \exp \left[\sum_{i=1}^N \theta_i^t x_i + \sum_{j>i} \theta_{ij}^t x_j x_i - \psi_t(\boldsymbol{\theta}_t) \right], \quad (1)$$

where $\psi_t(\boldsymbol{\theta}_t)$ is a log normalization term (a.k.a. a log partition function). This model is called spin-glass or Ising model in statistical mechanics, and contains $N + N(N+1)/2$ state variables ($\{\theta_i\}, \{\theta_{ij}\}$), also called natural parameters. By introducing the state vector $\boldsymbol{\theta}_t = (\theta_1, \dots, \theta_N, \theta_{1,2}, \dots, \theta_{N-1,N})'$, and the feature vector $\mathbf{F}(\mathbf{x}) = (x_1, \dots, x_N, x_{1,2}, \dots, x_{N-1,N})'$, an uncluttered expression of Eq. 1 is given as $p(\mathbf{x}|\boldsymbol{\theta}_t) = \exp[\boldsymbol{\theta}_t' \mathbf{F}(\mathbf{x}) - \psi_t(\boldsymbol{\theta}_t)]$. The log partition function is then given by

$$\psi_t(\boldsymbol{\theta}_t) = \log \sum_{\mathbf{x}} \exp[\boldsymbol{\theta}_t' \mathbf{F}(\mathbf{x})]. \quad (2)$$

In statistical mechanics it is known as the free energy of the system. It should also be noted that it specifies the probability that all neurons are simultaneously silent because $p(\mathbf{0}|\boldsymbol{\theta}_t) = \exp[-\psi_t(\boldsymbol{\theta}_t)]$. This model considers individual and pairwise activity of neurons. Hence, it will be referred to as the *pairwise observation model* in the following.

The state model assumes dynamics of the latent state $\boldsymbol{\theta}_t$ are described by a random walk

$$\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} + \boldsymbol{\xi}_t(\boldsymbol{\lambda}), \quad (3)$$

where $\boldsymbol{\xi}_t$ is a random vector drawn from a multivariate normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{Q})$, and $\mathbf{Q}$ is a diagonal covariance matrix. The diagonal of the inverse matrix $\mathbf{Q}^{-1}$ is given by a vector $\boldsymbol{\lambda}$ that determines precision of the noise for each element. For the initial time bin we assume the density is $p(\boldsymbol{\theta}_1) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Estimating the state-space model Given the data $\mathbf{X}^{1:T}$, our goal is to jointly estimate the posterior density of the latent states and the optimal noise precision $\boldsymbol{\lambda}$. By denoting hyperparameters of the model as $\mathbf{w} = (\boldsymbol{\lambda}, \boldsymbol{\mu}, \boldsymbol{\Sigma})$ the posterior density of the state process writes as

$$p(\boldsymbol{\theta}_{1:T}|\mathbf{X}^{1:T}, \mathbf{w}) = \frac{p(\mathbf{X}^{1:T}|\boldsymbol{\theta}_{1:T})p(\boldsymbol{\theta}_{1:T}; \mathbf{w})}{p(\mathbf{X}^{1:T}; \mathbf{w})}, \quad (4)$$

where the first component in the numerator is constructed from the observation model, and the second component from the state model. The posterior density depends on the choice of the parameters $\mathbf{w}$. The optimal $\mathbf{w}$ maximizes the marginal likelihood, a.k.a. evidence, that appears in the denominator in Eq. 4, given by

$$l(\mathbf{X}^{1:T}|\mathbf{w}) = p(\mathbf{X}^1|\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \prod_{t=2}^T p(\mathbf{X}^t|\mathbf{X}^{1:t-1}, \mathbf{w}). \quad (5)$$

This approach is called the empirical Bayes method. In this study, we optimize $\boldsymbol{\lambda}$ as described below while values for the hyperparameters $\boldsymbol{\mu}, \boldsymbol{\Sigma}$ of the initial distribution are fixed.

The optimization is achieved by an EM-algorithm combined with recursive Bayesian filtering/smoothing algorithms [35, 36]. In this approach, construction of the posterior density (Eq. 4, E-step) and optimization of the hyperparameters (M-step) are alternately performed until the marginal likelihood (Eq. 5) saturates. In order to update the hyperparameters to new values $\mathbf{w}^*$ from old values $\mathbf{w}$ in the M-step, we maximize a lower bound of the marginal likelihood. This bound is also called expected complete data log-likelihood, where the expectation is taken by the posterior density given by Eq. 4 with the old $\mathbf{w}$. This bound is written as

$$\begin{aligned}
l(\mathbf{X}^{1:T}|\mathbf{w}) &\geq \langle \log p(\mathbf{X}^{1:T}, \boldsymbol{\theta}_{1:T}|\mathbf{w}^*) \rangle_{\boldsymbol{\theta}_{1:T}|\mathbf{X}^{1:T}, \mathbf{w}} \\
&= \sum_{r=1}^R [\boldsymbol{\theta}_t' \mathbf{F}(\mathbf{X}^{t,r}) - \psi(\boldsymbol{\theta}_t)] - \frac{1}{2} \log |2\pi \boldsymbol{\Sigma}^*| \\
&\quad - \frac{1}{2} \langle (\boldsymbol{\theta}_1 - \boldsymbol{\mu}^*)' \boldsymbol{\Sigma}^{*-1} (\boldsymbol{\theta}_1 - \boldsymbol{\mu}^*) \rangle_{\boldsymbol{\theta}_{1:T}|\mathbf{X}^{1:T}, \mathbf{w}} \\
&\quad - \frac{T-1}{2} \log |2\pi \mathbf{Q}^*| - \frac{1}{2} \sum_{t=2}^T \langle (\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t-1})' \mathbf{Q}^{*-1} (\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t-1}) \rangle_{\boldsymbol{\theta}_{1:T}|\mathbf{X}^{1:T}, \mathbf{w}}.
\end{aligned} \tag{6}$$

Here $\langle \cdot \rangle_{\boldsymbol{\theta}_{1:T}|\mathbf{X}^{1:T}, \mathbf{w}}$ is the expectation by the posterior density of the state variables (Eq. 4). By considering the derivative of this equation w.r.t. the hyperparameters, we obtain their update rules. For the precision $\boldsymbol{\Lambda}^* (= \text{diag } \mathbf{Q}^{*-1})$ the updated value is computed using

$$\mathbf{Q}^* = \frac{1}{T-1} \sum_{t=2}^T \langle (\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t-1})(\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t-1})^T \rangle_{\boldsymbol{\theta}_{1:T}|\mathbf{X}^{1:T}, \mathbf{w}}. \tag{7}$$

Here the key step is to develop an algorithm that constructs the posterior density of Eq. 4. This is done by the forward and backward recursive Bayesian algorithms. Below we review this method followed by introduction of the approximations that make the method applicable to larger number of neurons.

Recursive estimation of dynamic neural interactions The estimation of the latent process is achieved by forward filtering and then backward smoothing algorithms. In the filtering algorithm, the state of population activity at time bin t is sequentially estimated using the data up to time t . This estimate is given by the recursive Bayesian formula

$$p(\boldsymbol{\theta}_t|\mathbf{X}^{1:t}, \mathbf{w}) = \frac{p(\mathbf{X}^t|\boldsymbol{\theta}_t)p(\boldsymbol{\theta}_t|\mathbf{X}^{1:t-1}, \mathbf{w})}{p(\mathbf{X}^t|\mathbf{X}^{1:t-1}, \mathbf{w})}. \tag{8}$$

where $p(\mathbf{X}^t|\boldsymbol{\theta}_t)$ is obtained from the observation model. The second term in the numerator $p(\boldsymbol{\theta}_t|\mathbf{X}^{1:t-1}, \mathbf{w})$ is called the one-step prediction density. It is computed using the state model and the filter density at the previous time bin via the Chapman-Kolmogorov equation,

$$p(\boldsymbol{\theta}_t|\mathbf{X}^{1:t-1}, \mathbf{w}) = \int p(\boldsymbol{\theta}_t|\boldsymbol{\theta}_{t-1}, \mathbf{w})p(\boldsymbol{\theta}_{t-1}|\mathbf{X}^{1:t-1}, \mathbf{w})d\boldsymbol{\theta}_{t-1}. \tag{9}$$

Thus the filter density (Eq. 8) can be recursively computed for $t = 2, \dots, T$ using Eq. 9, given observation and state models as well as an initial distribution of the one-step prediction density at time $t = 1$. Note that the initial one-step prediction density was specified as $p(\boldsymbol{\theta}_1) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. This distribution dictates the density of the state at the initial time step without observing neural activity.

The approximate nonlinear recursive formulae were developed by approximating the posterior density (Eq. 8) with a Gaussian distribution [37, 38]. Let us assume that the filter density at time $t-1$ is given by a Gaussian distribution with mean $\boldsymbol{\theta}_{t-1|t-1}$ and the covariance matrix $\mathbf{W}_{t-1|t-1}$. The subscript $t-1|t-1$ means the estimate at time $t-1$ (left) given the data up to time bin $t-1$ (right). Because the state model (Eq. 3) is also Gaussian the Chapman-Kolmogorov equation yields the one-step prediction density that is a Gaussian distribution with mean $\boldsymbol{\theta}_{t|t-1} = \boldsymbol{\theta}_{t-1|t-1}$ and

covariance $\mathbf{W}_{t|t-1} = \mathbf{W}_{t-1|t-1} + \mathbf{Q}$. The log posterior density (Eq. 8) then is written as

$$\begin{aligned} \log p(\boldsymbol{\theta}_t | \mathbf{X}^{1:t}, \mathbf{w}) &= \sum_{r=1}^R [\boldsymbol{\theta}_t' \mathbf{F}(\mathbf{X}^{t,r}) - \psi(\boldsymbol{\theta}_t)] \\ &\quad - \frac{1}{2}(\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t|t-1})' \mathbf{W}_{t|t-1}^{-1} (\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t|t-1}) + \text{const.} \end{aligned} \quad (10)$$

Here we approximate the posterior density by a Gaussian distribution (Laplace approximation). We identify the mean of this distribution with the MAP estimate:

$$\boldsymbol{\theta}_{t|t} = \text{argmax}_{\boldsymbol{\theta}_t} \log p(\boldsymbol{\theta}_t | \mathbf{X}^{1:t}, \mathbf{w}). \quad (11)$$

This solution is called a filter mean. It may be obtained by gradient ascent algorithms such as the conjugate gradient algorithm and the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm. These algorithms use the gradient

$$\frac{\partial \log p(\boldsymbol{\theta}_t | \mathbf{X}^{1:t}, \mathbf{w})}{\partial \boldsymbol{\theta}_t} = \sum_{r=1}^R [\mathbf{F}(\mathbf{X}^{t,r}) - \boldsymbol{\eta}_t] - \mathbf{W}_{t|t-1}^{-1} (\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t|t-1}). \quad (12)$$

Here the expectation parameters $\boldsymbol{\eta}_t$ are defined as

$$\boldsymbol{\eta}_t \equiv \frac{\partial \psi(\boldsymbol{\theta}_t)}{\partial \boldsymbol{\theta}_t} = \langle \mathbf{F}(\mathbf{x}) \rangle_{\boldsymbol{\theta}_t}, \quad (13)$$

where $\langle \mathbf{x} \rangle_{\boldsymbol{\theta}_t}$ is the expectation of $\mathbf{x}$ with respect to $p(\mathbf{x} | \boldsymbol{\theta}_t)$. This expectation needs to be computed repeatedly in the gradient algorithms. The covariance matrix of the approximated Gaussian distribution is computed from the Hessian of the log posterior evaluated at the MAP estimate:

$$\begin{aligned} \mathbf{W}_{t|t}^{-1} &= - \left. \frac{\partial^2 \log p(\boldsymbol{\theta}_t | \mathbf{X}^{1:t}, \mathbf{w})}{\partial \boldsymbol{\theta}_t \partial \boldsymbol{\theta}_t'} \right|_{\boldsymbol{\theta}_{t|t}} \\ &= R \mathbf{G}_t + \mathbf{W}_{t|t-1}^{-1}. \end{aligned} \quad (14)$$

$\mathbf{G}_t$ is the Fisher-information matrix:

$$\mathbf{G}_t \equiv \left. \frac{\partial \psi(\boldsymbol{\theta}_t)}{\partial \boldsymbol{\theta}_t \partial \boldsymbol{\theta}_t'} \right|_{\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t|t}} = \langle \mathbf{F}(\mathbf{x}) \mathbf{F}(\mathbf{x})' \rangle_{\boldsymbol{\theta}_{t|t}} - \langle \mathbf{F}(\mathbf{x}) \rangle_{\boldsymbol{\theta}_{t|t}} \langle \mathbf{F}(\mathbf{x}) \rangle_{\boldsymbol{\theta}_{t|t}}'. \quad (15)$$

The expectations are taken by $p(\mathbf{x} | \boldsymbol{\theta}_{t|t})$. Note that we initially assumed that the filter density at previous time step is a Gaussian distribution when computing the Chapman-Kolmogorov equation. In the next step, this assumption is fulfilled by this Laplace approximation. We additionally assumed that the initial distribution of the state variables is Gaussian. Thus we obtain an approximate nonlinear recursive filter that is consistent across the iterations.

Once the approximate filter density is constructed for $t = 1, \dots, T$, the backward smoothing algorithm is applied to obtain the smoothed posterior density of the state variable at time t [38, 39],

$$p(\boldsymbol{\theta}_t | \mathbf{X}^{1:T}, \mathbf{w}) = p(\boldsymbol{\theta}_t | \mathbf{X}^{1:t}, \mathbf{w}) \int \frac{p(\boldsymbol{\theta}_{t+1} | \mathbf{X}^{1:T}, \mathbf{w}) p(\boldsymbol{\theta}_{t+1} | \boldsymbol{\theta}_t, \mathbf{w})}{p(\boldsymbol{\theta}_{t+1} | \mathbf{X}^{1:T}, \mathbf{w})} d\boldsymbol{\theta}_{t+1}. \quad (16)$$

for $t = T, \dots, 1$. In practice, the following fixed interval smoothing algorithm [38] provides the smoothed MAP estimate $\boldsymbol{\theta}_{t|T}$ and smoothed covariance $\mathbf{W}_{t|T}$ of the posterior distribution

$$\boldsymbol{\theta}_{t|T} = \boldsymbol{\theta}_{t|t} + \mathbf{A}_t (\boldsymbol{\theta}_{t+1|T} - \boldsymbol{\theta}_{t+1|t}), \quad (17)$$

$$\mathbf{W}_{t|T} = \mathbf{W}_{t|t} + \mathbf{A}_t (\mathbf{W}_{t+1|T} - \mathbf{W}_{t+1|t}) \mathbf{A}_t', \quad (18)$$

where $\mathbf{A}_t = \mathbf{W}_{t|t} \mathbf{W}_{t+1|t}^{-1}$. In addition, the posterior covariance matrix between state variables at time t and $t-1$ is obtained as $\mathbf{W}_{t-1,t|T} = \mathbf{A}_{t-1} \mathbf{W}_{t|T}$ [40]. This procedure constructs the smoother posterior density of the latent process (Eq. 4) by approximating it as a Gaussian process of length $N(N+1)/2 \times T$ with mean $(\boldsymbol{\theta}'_{1|T}, \boldsymbol{\theta}'_{2|T}, \dots, \boldsymbol{\theta}'_{T|T})$ and a banded diagonal covariance matrix whose block diagonal is given by $\mathbf{W}_{t|T}$ (for $t = 1, \dots, T$), and block-off diagonals are given by $\mathbf{W}_{t-1,t|T}$ (for $t = 2, \dots, T$).

Approximation methods for large-scale analysis

Approximate estimate of filter mean by pseudolikelihood method

To obtain the filter estimate using iterative gradient ascent methods the gradient (Eq. 12) needs to be evaluated at each iteration. This requires computation of the expectations (Eq. 13) by summing over all 2^N states the network can realize. For a large network size N , this is infeasible. Thus the method introduced in the previous subsection was limited to $N \leq 15$. However, the *pseudolikelihood* method [23, 41, 42] has been shown to accurately estimate the interactions without requiring evaluation of the expectations. Here we incorporate it into the sequential Bayesian estimation framework.

The pseudolikelihood approximates the likelihood of the joint activity of neurons by a product of conditional likelihoods of each neuron given the activity of the others. Let the activity of neurons except neuron n be $\mathbf{x}_{\setminus n} = (x_1, \dots, x_{n-1}, x_{n+1}, \dots, x_N)$; and $f_t^n(\mathbf{x}_{\setminus n}) = \boldsymbol{\theta}_t' \mathbf{F}(x_n = 1, \mathbf{x}_{\setminus n})$. Then the pseudolikelihood is given by

$$\prod_{r=1}^R \tilde{p}(\mathbf{X}^{t,r} | \boldsymbol{\theta}_t) = \prod_{r=1}^R \prod_{n=1}^N p(X_n^{t,r} | \mathbf{X}_{\setminus n}^{t,r}, \boldsymbol{\theta}_t) = \prod_{r=1}^R \prod_{n=1}^N \frac{\exp(X_n^{t,r} f_t^n(\mathbf{X}_{\setminus n}^{t,r}))}{1 + \exp(f_t^n(\mathbf{X}_{\setminus n}^{t,r}))}. \quad (19)$$

Note that the log partition function does not appear in Eq. 19. Replacing the likelihood in Eq. 8 with Eq. 19 yields

$$\begin{aligned} \log p(\boldsymbol{\theta}_t | \mathbf{X}^{1:T}, \mathbf{w}) &\approx \sum_{r=1}^R \sum_{n=1}^N \left[X_n^{t,r} f_t^n(\mathbf{X}_{\setminus n}^{t,r}) - \log \left(1 + \exp(f_t^n(\mathbf{X}_{\setminus n}^{t,r})) \right) \right] \\ &\quad - \frac{1}{2} (\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t|t-1})' \mathbf{W}_{t|t-1}^{-1} (\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t|t-1}) + \text{const.} \end{aligned} \quad (20)$$

The derivative of this approximated filter density results in

$$\frac{\partial \log p(\boldsymbol{\theta}_t | \mathbf{X}^{1:T}, \mathbf{w})}{\partial \boldsymbol{\theta}_t} \approx \sum_{r=1}^R \sum_{n=1}^N \left[(X_n^{t,r} - \tilde{\eta}_n^{t,r}) \frac{\partial f_t^n(\mathbf{X}_{\setminus n}^{t,r})}{\partial \boldsymbol{\theta}_t} \right] - \mathbf{W}_{t|t-1}^{-1} (\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t|t-1}), \quad (21)$$

where $\tilde{\eta}_n^{t,r} = \langle x_n^t | \mathbf{X}_{\setminus n}^{t,r} \rangle_{\boldsymbol{\theta}_t}$, i.e., the expectation of x_n^t being 1 given the activity of the other neurons. Using this gradient in the same gradient ascent algorithms as before we obtain the approximate mean $\boldsymbol{\theta}_{t|t}$ of the filter density.

Approximation of the filter covariance

The pseudolikelihood can provide the approximate mode of the filter density (Eq. 11). However, to perform the sequential estimation, we need in addition the filter covariance matrix (Eq. 14). This requires to compute the Fisher information matrix (Eq. 15, i.e., the Hessian of the observation model at the filter mean $\boldsymbol{\theta}_{t|t}$). To compute the Fisher information matrix, not only the first and second order but also the third and fourth order expectation parameters need to be evaluated at the filter mean parameters. In order to avoid computing the higher-order expectation parameters and to reduce the computational cost of the matrix inversion, we approximate it by a diagonal matrix. This matrix is composed of the first and second order expectation parameters ($\{\eta_i^{t|t}\}, \{\eta_{ij}^{t|t}\}$), where the expectations parameters are defined as $\eta_i^{t|t} \equiv \langle x_i \rangle_{\boldsymbol{\theta}_{t|t}}$ and $\eta_{ij}^{t|t} \equiv \langle x_i x_j \rangle_{\boldsymbol{\theta}_{t|t}}$. Two different approximation methods are tested to obtain these marginals. One is the mean-field *Thouless-Anderson-Palmer* (TAP) approach [43] and the other the *Bethe approximation* [44].

TAP approximation The TAP approximation of the expectation parameters $\boldsymbol{\eta}_{t|t}$ given the natural parameters $\boldsymbol{\theta}_{t|t}$ (*forward-problem*) can be derived in multiple ways [10, 45], but here we follow [46, 47] that use the so-called Plefka expansion. The following formulae and their derivation are revised for binary variables $x_i \in \{0, 1\}$ instead of $\{-1, 1\}$. See Supporting Information S1 for more details. The method constructs a new free energy as a function of the mixture coordinates ($\{\eta_i^{t|t}\}, \{\eta_{ij}^{t|t}\}$) by the Legendre transformation of the log partition function ψ_t as $\sum_{i=1}^N \theta_i^{t|t} \eta_i^{t|t} - \psi_t$.

Then this function is approximated by a second-order expansion around the independent model assuming weak pairwise interactions. This results in the approximate log partition function,

$$\begin{aligned} \psi_t \approx & \sum_{i=1}^N \theta_i^{t|t} \eta_i^{t|t} - \sum_{i=1}^N \{ \eta_i^{t|t} \log \eta_i^{t|t} + (1 - \eta_i^{t|t}) \log(1 - \eta_i^{t|t}) \} + \frac{1}{2} \sum_{j \neq i} \theta_{ij}^{t|t} \eta_i^{t|t} \eta_j^{t|t} \\ & + \frac{1}{8} \sum_{j \neq i} (\theta_{ij}^{t|t})^2 \{ \eta_i^{t|t} - (\eta_i^{t|t})^2 \} \{ \eta_j^{t|t} - (\eta_j^{t|t})^2 \}. \end{aligned} \quad (22)$$

Here we extended the definition of interaction parameters as $\theta_{ii}^{t|t} = 0$ and $\theta_{ij}^{t|t} = \theta_{ji}^{t|t}$. At the independent model the values for the expectations can be computed and the expansion yields correction terms for the non-zero $\theta_{ij}^{t|t}$ s. Since the derivative of the new free energy based on the mixture coordinates w.r.t. $\{\eta_i^{t|t}\}$ yields the first order parameters $\{\theta_i^{t|t}\}$, we obtain the following self-consistent equations:

$$\theta_i^{t|t} = \log \left(\frac{\eta_i^{t|t}}{1 - \eta_i^{t|t}} \right) - \sum_{j \neq i} \theta_{ij}^{t|t} \eta_j - \frac{1}{2} \sum_{j \neq i} (\theta_{ij}^{t|t})^2 \left(\frac{1}{2} - \eta_i^{t|t} \right) \{ \eta_j^{t|t} - (\eta_j^{t|t})^2 \}, \quad (23)$$

for $i, j = 1, \dots, N$. Solving this equations yields the first order expectations which can be used to estimate the log partition function (Eq. 22).

Further from the relation $\frac{\partial \theta_i^{t|t}}{\partial \eta_j^{t|t}} = [\mathbf{G}_t^{-1}]_{ij}$, we obtain

$$[\mathbf{G}_t^{-1}]_{ij} = \left(\frac{1}{\eta_i^{t|t} (1 - \eta_i^{t|t})} \right) \delta_{ij} - \theta_{ij}^{t|t} - (\theta_{ij}^{t|t})^2 \left(\frac{1}{2} - \eta_i^{t|t} \right) \left(\frac{1}{2} - \eta_j^{t|t} \right). \quad (24)$$

Here δ_{ij} is the delta function, which is 1 for $i = j$ and 0 otherwise. To obtain the second order expectation parameters, we calculate and then invert the $N \times N$ matrix obtained by Eq. 24, and approximate it as the Fisher information matrix for $\{\theta_i\}$ given in Eq. 15 to obtain the second order expectation parameters by $\eta_{ij}^{t|t} = [\mathbf{G}_t]_{ij} + \eta_i^{t|t} \eta_j^{t|t}$ [46].

Bethe approximation The Bethe approach approximates a probability distribution by assuming that it factorizes into its pairwise marginals. Hence, the approximated joint distribution writes as

$$p(\mathbf{x}|\boldsymbol{\theta}_{t|t}) \approx \frac{\prod_{i,j>i} q_t(x_i, x_j)}{\prod_i q_t(x_i)^{(N-1)}} := q_t(\mathbf{x}), \quad (25)$$

where q are so-called *beliefs* [48] that approximate the marginals of the underlying distribution p . Note that for any acyclic graph this yields the true joint distribution. However, here the observation model (Eq. 1) is a fully connected graph and hence the Bethe approximation ignores all cycles. Realizing that the beliefs have to fulfill some constraints one can write the problem as a Lagrangian that has to be minimized. This allows to derive a dual representation of the marginals (in terms of the Lagrangian multipliers), which in turn allows to derive messages that are sent from one belief to another. Propagating this beliefs through the Markov field yields the belief propagation algorithm (BP) [44]. While BP is relatively fast, it is not guaranteed to converge to a unique solution. This guarantee is provided by the alternative concave-convex procedure (CCCP) [49]. CCCP also starts from the same Lagrangian, but updates the beliefs and Lagrangian multipliers in an alternating manner. This more strict procedure comes with the disadvantage that it is much slower than BP. Therefore, here the two algorithms are combined to a *hybrid method* in the following, where BP is utilized primarily and the algorithm falls back to CCCP, when BP does not converge. For more details on the Bethe approximation see Supporting Information S2.

The estimation of the log partition function for the Bethe approximation is simply computed by the negative logarithm of the approximated probability (Eq. 25) that all neurons are silent, i.e

$$\psi_t \approx -\log q(\mathbf{0}|\boldsymbol{\theta}_{t|t}). \quad (26)$$

Because TAP as well as Bethe approximation provide an estimate of the log partition function ψ_t one is able to evaluate the approximation of the marginal likelihood (Eq. 6), and the EM-algorithm for the state-space model can be run until it converges.

Results

Model fit to simulated data In the following subsections we demonstrate the fit of the state-space model of neural population activity to artificially generated data of 30 neurons with dynamic couplings for $T = 500$ time bins. To be able to compare it to the ground truth we construct 2 populations each consisting of 15 neurons. Individual parameters $\theta_{1:T}$ of the underlying submodels are generated as smooth Gaussian processes, where the mean for the first order parameters $\{\theta_i^t\}_{i=1,\dots,N}$ increases at $t = 100$ and then decreases more slowly shortly after that. The interaction parameters $\{\theta_{ij}^t\}$ are generated by Gaussian processes whose mean is fixed at 0. In total, 200 trials are sampled from this generative model. Note that the sampled individual parameters differ and vary over time although we use homogeneous means. The increase of the mean for $\{\theta_i\}_{i=1,\dots,N}$ increases spiking probability followed by a decrease back to baseline (Fig. 1 A). In the resulting spike data, neurons spike with time averaged probabilities ranging from 0.13 up to 0.25. Supposing the bin width $\Delta = 10$ ms these are in a physiologically reasonable range. This exemplary scenario may mimic a population that independently receives an external input elicited by e.g. a sensory stimulus. For details of the generation of the data see Supporting Information S3.

Next we fit the state-space model of neural population activity to the generated data with the combination of pseudolikelihood and Bethe approximation. This combination is chosen for the demonstration because it provides the best estimates of the underlying model as we will assess in the later section. Top panel of Fig. 1B shows snapshots of the smoothed estimates of dynamic interactions ($\theta_{ij}^{t|T}$) at different time points ($t = 50, 150, 300$). The color of the nodes indicate the smoothed estimates of the first order parameters ($\theta_i^{t|T}$). Visual inspection of the fitted networks suffices to identify that there are 2 independent subpopulations of correlated neurons (one is the upper half, and the other the lower half). To check whether these changes match those of the underlying generative model, confidence intervals of three fitted couplings θ_{ij} are compared with their underlying values (Fig. 1B Bottom). The fit follows the dynamics, and correctly identifies the parameter that is constantly 0 (the lowest panel).

Estimating macroscopic properties of the network One of the main motives to model joint activities of a large population of neurons is to assess macroscopic properties of the network in a time dependent manner with confidence intervals. The macroscopic measures obtained for this example are shown in Fig. 1C, and in the following we introduce them one by one.

The first and simplest macroscopic property shown in the top left panel of Fig. 1C is the probability of spiking in a network (population spike rate). We define it as

$$p_{\text{spike}}(t) = \frac{1}{N} \sum_{i=1}^N \eta_i^t, \quad (27)$$

where η_i^t is the spike rate of i th neuron at time t . Using $\eta_i^t = \eta_i^{t|T}$, the method gives a smooth estimate of the empirical rate obtained from the data (Fig. 1A Bottom). The shaded area in the panel indicates the 99% confidence interval of the population spike rate obtained by sampling the natural parameters from the smoothed posterior density at each bin. The underlying spike probability for $N = 30$ neurons is obtained by calculating the marginals $\{\eta_i^t\}_{i=1,\dots,N}$ independently for each subpopulation and averaging over all neurons.

Next from the state-space model of neural population activity one can estimate the probability of simultaneous silence (i.e. the probability that no neuron elicits a spike, Fig. 1C bottom left)

$$p_{\text{silence}}(t) = \exp(-\psi_t). \quad (28)$$

The approximation methods allowed us to evaluate the log partition function ψ_t (Eq. 22 and 26. Here we used smoothed estimates to compute the log partition function). Thus we immediately obtain the probability of simultaneous silence. The simultaneous silence of $N = 30$ neurons is obtained as multiplication of the silence probabilities of the two subpopulations.

The entropy of the network (i.e., expectation of the information content, $\langle -\log p(\mathbf{x}|\theta_t) \rangle_{\theta_t}$) can be also calculated from the model as

$$S(t) = -\theta'_t \eta_t + \psi_t. \quad (29)$$

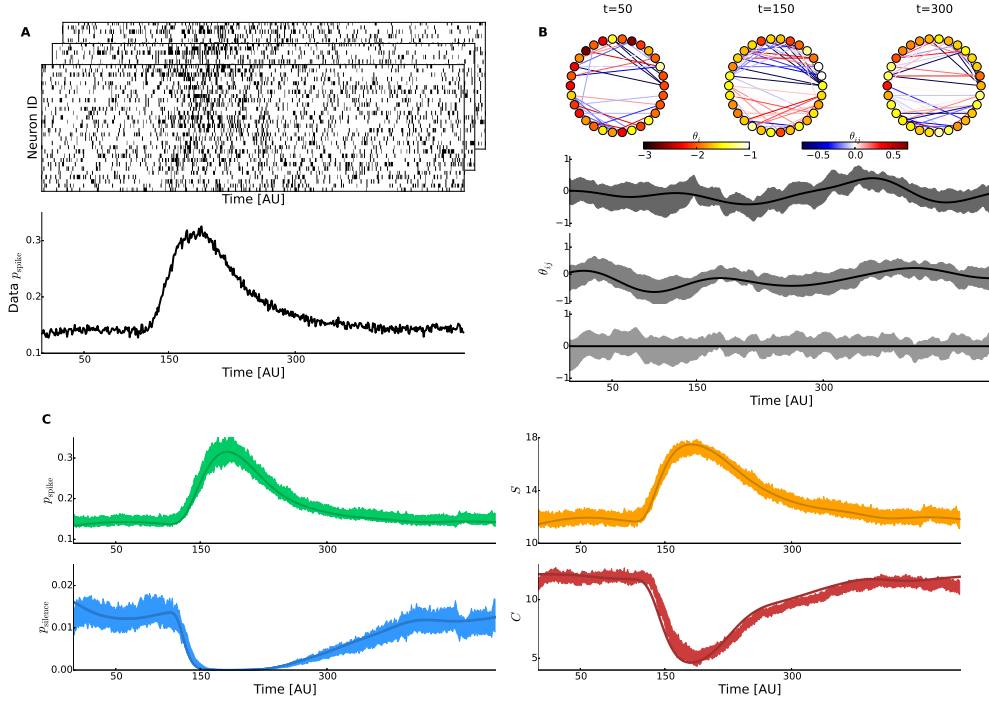


Figure 1: Approximate inference of dynamic neural interactions and macroscopic network properties. Simulated spike data of 30 neurons. **A** Top: Simultaneous spiking activity of 30 neurons that are repeatedly simulated 200 times (Here only 3 repetitions are visualized). The data was sampled from a time-dependent model of a neural population (Eq. 1). The time-varying parameters were chosen such that neurons' spike probability resembles evoked activity in response to stimulus presentation to an animal. The neural interactions were assumed to smoothly change irrespective of the firing rates. See the main text for details. Bottom: Empirical spike probability over time, averaged over trials and neurons. **B** Top: Estimated network states at $t = 50, 150, 300$ by the pseudolikelihood Bethe approximation method (see main text). Neurons are represented by nodes whose colors respectively indicate a value of the smoothed estimate of θ_i^t (for $i = 1, \dots, 30$). Links are color-coded according to estimated strength of the interaction θ_{ij}^t between connected nodes (positive or negative interactions are marked in red or blue, respectively). Only significant edges are displayed, where the corresponding θ_{ij}^t has a 99% confidence interval that does not include 0. Bottom: Dynamics of 3 exemplary interaction parameters, θ_{ij} . The lines denote the ground truth from which the binary data were sampled. The shaded areas are 99% credible intervals. **C** Estimated population rate (top left). Probability that all neurons are silent (bottom left). Entropy (top right) and sensitivity (bottom right) of the neural population. In all panels, shaded areas indicate 1% and 99% quantiles obtained by resampling θ from the fitted smoothed distribution. Solid lines represent ground truth computed from the underlying network model.

Estimation of this information theoretic measure allows us to quantify the amount of interactions in the network by comparing the pairwise model to the independent one (see following analysis and Eq. 34). Since it is extensive quantity, the entropy of $N = 30$ neurons is obtained by addition of the entropy from the two independent subpopulations. The entropy increased while the individual activity rates of neurons are also increased (Fig. 1C top right).

The last measure shown in the bottom right panel of Fig. 1C is the heat capacity, or sensitivity, of the system. It is the variance of information content: $C(t) = \langle \log p(\mathbf{x}|\theta_t)^2 \rangle_{\theta_t} - \langle \log p(\mathbf{x}|\theta_t) \rangle_{\theta_t}^2$, where the brackets indicate expectation by $p(\mathbf{x}|\theta_t)$. It is also the variance of the Hamiltonian $-\theta_t' \mathbf{F}(\mathbf{x})$. Thus it can be obtained by introducing a nominal dual parameter β to the Hamiltonian in the model,

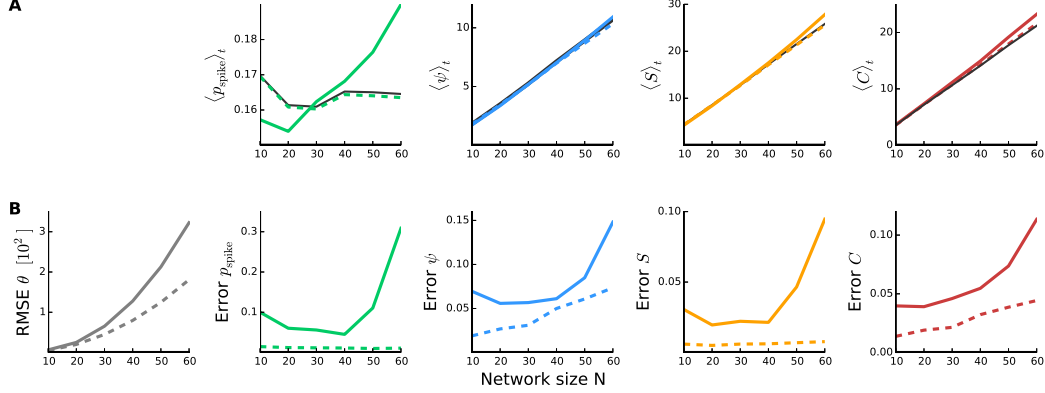


Figure 2: **Approximation error and network size** Networks consisting of subpopulations with 10 neurons are constructed following the same procedure as in Fig 1. **A:** The average value of the macroscopic properties over time as a function of network size. Black line is the true value, while colored lines show the estimated ones (solid line fit with 200 trials and dashed with 500 trials) **B:** The corresponding normalized error (only for θ the RMSE) for 200 trials (solid) and 500 trials (dashed).

assuming that it was 1 for real data. The log partition function of the augmented model is

$$\psi_t(\beta) = \log \sum_{\mathbf{x}} \exp(\beta \boldsymbol{\theta}'_t \mathbf{F}(\mathbf{x})). \quad (30)$$

The variance of Hamiltonian is given as the Fisher information w.r.t. β , i.e., the second derivative of the log partition function. This allows us to use the approximate ψ_t to assess the heat capacity. Then we further approximate the second derivative by its discrete version

$$C(t) = \left. \frac{\partial^2 \psi_t}{\partial \beta^2} \right|_{\beta=1} \approx \frac{\psi_t(1 + \epsilon) - 2\psi_t(1) + \psi_t(1 - \epsilon)}{\epsilon^2}, \quad (31)$$

and ϵ was chosen to be 10^{-3} . Sensitivity measures how drastically the total network changes its behavior due to subtle changes in its configuration (i.e., to changes of the $\boldsymbol{\theta}$ parameters). Networks with higher sensitivity are more responsive to changes than those with lower sensitivity. The heat capacity is also an extensive quantity. For the simulated data, the heat capacity decreased while activity rates of neurons are increased (Fig. 1C bottom right).

Assessment of fitting error with different network sizes and amount of data Next we examine the goodness-of-fit of the model fitted by the pseudolikelihood and Bethe approximation methods. In particular, we ask how the fitting performance changes with increasing network size. For this reason 6 dynamic models for populations of 10 neurons are generated as described previously (500 time bins, 200 trials). We then construct smaller or larger populations by concatenating the independent groups. The model is then fitted by the pseudolikelihood and Bethe approximation methods to the first subnetwork, then two subnetworks, and so on, until we fit the model to a network containing 60 neurons composed of 6 independent groups. We obtain estimates of the macroscopic measures from the MAP estimates of the model parameters at each time bin. Figure 2A shows values of these measures averaged over time. The results show extensive properties of macroscopic measures (except for the population spike rate), and that the estimates may deviate for larger number of neurons.

To assess quality of the fit, first the root mean squared error for the natural parameters averaged across time bins is calculated

$$\text{RMSE}(\boldsymbol{\theta}_{t|T}) = \sqrt{\frac{1}{T} \sum_{t=1}^T (\boldsymbol{\theta}_{t|T} - \boldsymbol{\theta}_t)^2}, \quad (32)$$

where $\theta_{t|T}$ is the smoothed estimate of the underlying model θ_t . For the data sets with 200 trials, the RMSE increases superlinearly with network size (Fig. 2 **B** Left). Furthermore, the error for the macroscopic measures is assessed by

$$\text{Error}[f(\hat{\theta})] = \frac{\text{RMSE}(f(\theta_{t|T}))}{\frac{1}{T} \sum_{t=1}^T f(\theta_{t|T})}, \quad (33)$$

where $f(\hat{\theta})$ is any function of the macroscopic measures. The RMSE is defined similarly to Eq. 32 while substituting the parameters θ by function f . These errors also increase as the network size increases (Fig. 2**B**).

To understand whether these errors increase primarily due to the approximation methods used for the fit or because of the finite amount of data, the fit is repeated but now to spiking data with 500 trials. The error of the fit is reduced particularly for larger network size (Fig. 2**A**, **B** dashed lines) suggesting that the limited amount of data is mainly responsible for the estimation error.

Comparison between Bethe and TAP approximation To this end, only the Bethe approximation are used in combination with the pseudolikelihood to fit the model approximately. However, as discussed previously, the TAP approximation constitutes a potential alternative. To assess the quality of both approximations, one of the two subpopulations used for Fig. 1 was investigated (15 neurons, 500 time bins, 200 trials). The smaller network is considered because it allows to fit the model exactly without the approximation methods for the large-scale analysis. Here the exact method refers to the method in which the expectation parameters are calculated exactly at the gradient search for the MAP estimates of model parameters. It should be noted that we approximate the posterior density by the Gaussian distribution even for the ‘exact method’ in the recursive Bayesian algorithm. Comparison of the approximation methods with the exact method determines the error that is caused by the approximation methods and not by the finite amount of data.

First, investigation of three exemplary time points (Fig. 3**A**) reveals that both the pseudolikelihood-Bethe and the pseudolikelihood-TAP approximation recover the underlying parameters. We examine the error across time bins by the RMSE. Comparing RMSE of the approximation results with exact fit (Fig. 3**B**) demonstrates that the approximations perform worse (RMSE increased by $\sim 20\%$), but are both in the same range.

Since both, Bethe and TAP, provide an approximation for the log partition function ψ_t (Eq. 22 and 26), we assess their performance for the same data as in Fig. 3. The time evolution of simultaneous silence (directly linked to ψ by Eq. 28) is recovered by exact, Bethe, and TAP (Fig. 4 **A**). The results show that the TAP approximation slightly overestimated the probability in this example. This is also reflected in the RMSE (Fig. 4 **B**), where the Bethe approximation performs better than the TAP method even though the RMSE for the Bethe approximation increases compared to the exact method. Another important disadvantage of the TAP approximation is that the system of non-linear equations occasionally could not be solved. This happens more frequently when fitting larger networks and/or networks with stronger interactions. Therefore, for fitting the state-space model of population activity we suggest that the pseudolikelihood-Bethe approximation exhibits better performance, and we will use it again for the following analysis.

Dynamic network inference from V4 spiking data of behaving monkey We now apply the approximate inference method to analysis of monkey V4 neurons recorded while the animal performed repeatedly the following behavioral task. Each trial began when the monkey fixated its gaze within 1 degree of a centrally-positioned dot on a computer screen. After 150 ms, a drifting sinusoidal grating was presented for 2 s in the receptive field area of the neuronal population that was recorded, at which time the grating stimulus disappeared and the fixation point moved to a new, randomly chosen location on the screen, and the animal made an eye movement to fixate on the new location. Data epochs from 500 ms prior to grating stimulus onset until 500 ms after stimulus offset were extracted from the continuous recording for analysis. The spiking data obtained by micro-electrode recordings includes 112 single and multi units identified by their distinct wave forms. The experiment was performed at the University of Pittsburgh. All experimental procedures were approved by the University of Pittsburgh Institutional Animal Care and Use Committee, and were performed in accordance with the United States’ National Institutes of Health (NIH) *Guide for the Care and Use*

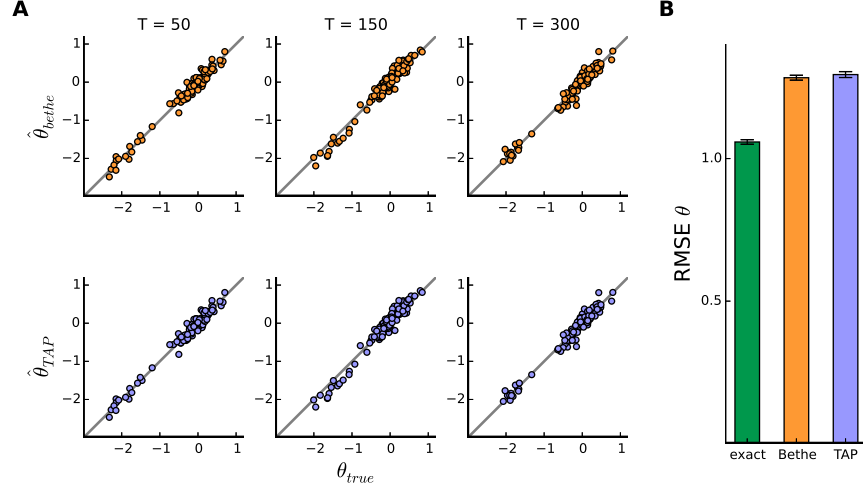


Figure 3: **Comparison of the Bethe and TAP approximation.** Simulated data of 15 neurons over 500 time bins (200 trials). **A** Top: Ground truth θ_t vs. its smoothed estimate by pseudolikelihood Bethe approximation at three different time points ($t = 50, 150, 300$). Bottom: The same as above obtained with pseudolikelihood TAP approximation. **B**: The RMSE between the true model parameter θ_t and its smoothed estimate by the exact inference, pseudolikelihood TAP, or pseudolikelihood Bethe approximation. The bar height and error bars indicate the mean and standard deviation from 10 realizations of data, each sampled from the same underlying parameters (generated as in Fig 1).

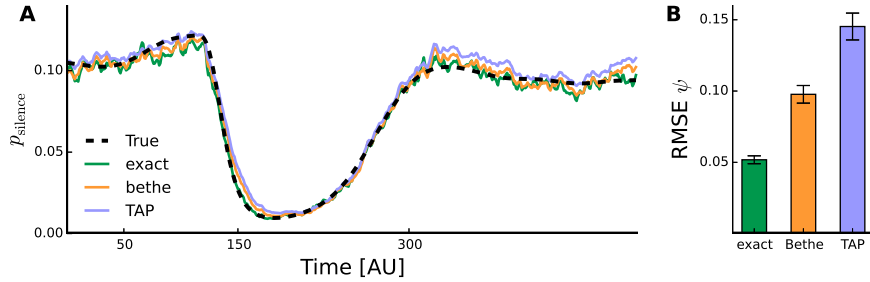


Figure 4: **Time-varying probability of simultaneous silence.** Results for a network of 15 neurons estimated by different approximation methods. **A** The probability of simultaneous silence ($p_{\text{silence}}(t) = \exp(-\psi_t)$) as a function of time. The pseudolikelihood Bethe (orange) and pseudolikelihood TAP (lavender) method estimate the underlying value with sufficient accuracy (dashed black). For comparison, an estimate by the exact method (green) is shown. **B** The RMSE between the approximate and true free energy ψ_t . The underlying model parameters are the same as in Fig 3.

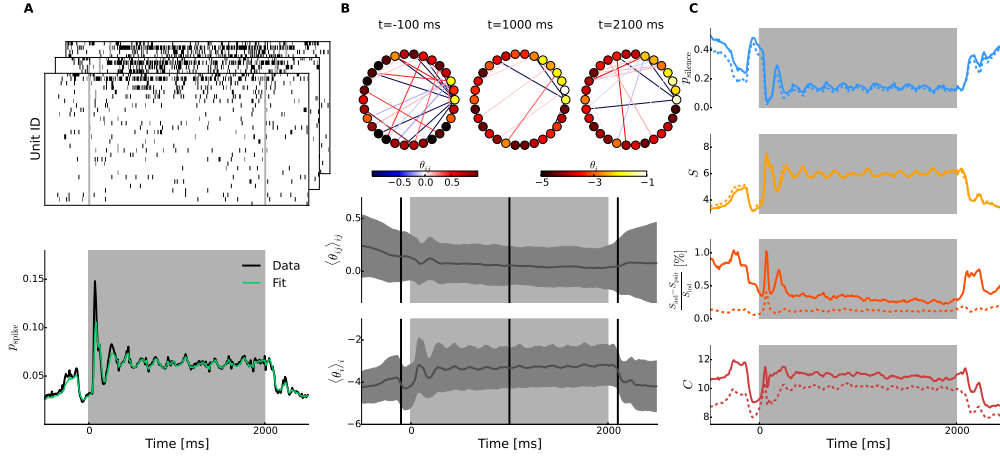


Figure 5: Dynamic network inference from monkey V4 data. In the paradigm a 90° grating on a screen was presented to the monkey for 2s (light gray shaded areas). 1004 trials were recorded and spike trains were obtained with bin width of 10ms. **A** Top: Exemplary spiking data. Bottom: Empirical probability (black) of observing a spike over time and spike probability of the fitted model (green). **B** Top: The fitted network at three different time points, before, during, and after stimulation. Edges with significantly non-zero θ_{ij} are displayed (as in Fig. 1). Bottom: The mean of θ_i 's and θ_{ij} 's of the smoothed MAP estimates (dark gray line). The shaded area is the mean plus/minus the standard deviation. **C** Macroscopic measures of the network over time obtained from the smoothed MAP estimates of the model (solid lines). The dashed lines are obtained from smoothed model fit to trial shuffled data.

of Laboratory Animals. For details on experimental setup, recording and unit identification see [50]. The recorded units are tested for across-trial stationarity (which is the assumption of the model). The mean firing rates for each trial are standardized and if more than 5% of the trials were outside the 95% confidence interval the unit is excluded. From the remaining units, 30 units with the highest firing rates are selected for the analysis. To obtain the binary data, the spike trains are discretized into time bins with $\Delta = 10$ ms resulting into 300 time bins over the course of the trial. Exemplary data are displayed in Fig. 5A Top.

After the data are preprocessed, the network dynamics of the 30 units during the task period are analyzed by the state-space model for the neural population activity. Inference is done by using the pseudolikelihood-Bethe approximation. The result of the fit is displayed in Fig. 5B. First, we observe stimulus locked oscillations in the population firing rate that are also captured by the model (Fig. 5A Bottom). The average of the estimated natural parameters (Fig. 5B Bottom) show that these oscillations are explained by the first order parameters ($\{\theta_i^{t|T}\}_{i=1,\dots,30}$). Investigation of the network states before, during, and after the stimulus (Fig. 5B Top) reveals that the interactions $\{\theta_{ij}^{t|T}\}$ are altered over time. This is also reflected in an average over the all pairwise interactions (Fig. 5B Center), where the mean decreases during the stimulus presentation as well as the standard deviation. Thus neurons are likely to decorrelate during the stimulus presentation whereas the population rate increases and oscillates at the same time.

Similarly to the analysis of artificial data (Fig. 1), we measure the macroscopic properties of the fitted model over the task period (Fig. 5C). Here to test the contribution of interactions in the recorded data, the model is once again fitted to trial shuffled data [17], which should destroy all correlations among units that do not occur due to chance. Comparison of the macroscopic measures between the models fitted to the original data and to the trial shuffled data shows how interactions among units alter the results. In the following, we will refer to the two models as ‘actual’ and ‘trial shuffled’ model.

The probability of simultaneous silence shows again the stimulus locked oscillations, and decreases during the stimulus period. The difference between the actual and trial shuffled model before the stimulus is larger than during and after the stimulus, suggesting that positive interactions contributed to increasing the silence probability in particular before and after the stimulus period. The entropy shows a strong increase ($\sim 1/3$) during the stimulus period. This is reasonable because we observe an increase in activity rates and a decrease in correlations - both effects should result in an increase in entropy. Next, we examine how much of the entropy is explained by the interactions among the neurons. To do so, at each time point we calculate a corresponding independent model by projecting the fitted interaction model to the independent model (i.e. the model with the same individual firing rates η_i but with all $\theta_{ij} = 0$). The entropy of the independent model S_{ind} should always be larger than S_{pair} , the entropy of the model with interactions. Hence, a fraction of entropy explained by the interactions can be calculated as

$$\frac{S_{ind} - S_{pair}}{S_{ind}}. \quad (34)$$

In general, contribution of interactions to the entropy is small for these data ($\leq 1\%$). However, during stimulus the contribution is less than half compared to the period before the stimulus. Only in the beginning of the stimulus presentation, two peaks can be observed. We thus conclude that neurons decorrelated during the stimulus presentation. The decorrelation observed during the stimulus period is successfully dissociated from the oscillatory population activity: Previously observed oscillations are absent in this measure of interactions. This result is important because ignoring such firing rate dynamics often leads to erroneous detection of positive correlations among neurons. Last, the sensitivity (heat capacity) of the network over time is obtained. While for the artificial data in Fig. 1 the sensitivity showed a drastic decrease, such reduction is not observed in the V4 data. The sensitivity of the network is maintained at approximately the same value before and during the stimulus period. This is interesting since we already observed that before and during the stimulus the network seems to be in two qualitatively different states (low vs. high firing rate and strong vs. weak interactions). After stimulus presentation the sensitivity drops. Overall, neural interactions contribute to have higher sensitivity (the solid line is higher than the dashed line).

Discussion

This study provides approximate inference methods for simultaneously estimating neural interactions of a large number of neurons, and quantifying macroscopic properties of the network in a time-resolved manner. Performance of these methods were assessed by using simulated parallel spike sequences, and utility of the proposed approach was demonstrated by revealing dynamic decorrelation of V4 neurons along with their oscillatory rate activity and maintained susceptibility during stimulus presentations.

Accurate assessment of correlated population activity in ongoing and evoked activity is a key to understand the underlying biological mechanisms and their coding principles. It is critical to model time-dependent firing rates to correctly assess neural interactions. If we apply a stationary model of neural interactions to independent neurons with varying firing rates, we may erroneously observe excess of correlations [16–18, 51]. Such an apparent issue of a stationary model can introduce considerable confusion in search of fundamental coding principles of neurons. For example, it was proposed that learned correlations among V1 neurons of adult ferret, as assessed by a stationary model of the population activity, adapted to natural environment because neural correlations during spontaneous and evoked activities progressively become closer to each other along developmental stages [52]. However, it was pointed out that the increased similarity of correlations might be explained by similar population rate dynamics during distinct cortical states [53]. Although our model does not contain homogeneous higher-order interactions that the population rate model imposes, the proposed state-space model may provide an alternative view on the learned correlations by explaining away the effect of time-dependent rates of individual neurons. Indeed we found a small contribution of neural correlations in characterizing activity of V4 neurons (Fig. 5C 3rd panel) partly because joint activity of neurons were explained by covariations of activity rates of individual neurons seen as the oscillatory population activity (Fig. 5A Bottom). Moreover, to investigate learning-induced neural correlations, one may apply this state-space model directly to chronically recorded data (see e.g., [54]), and trace modification in neural interactions during learning.

In this study, we employed the classical pseudolikelihood method to perform MAP estimation of interactions (i.e., natural parameters) without computing the partition function. For the inverse problem without the prior, we may use alternative approximation methods such as TAP and Bethe approximation, and further state-of-the-art methods such as the Sessak-Monasson [9], minimum-probability-flow [12], and adaptive-cluster expansion [14] method. However, here we chose the pseudolikelihood method because it was not trivial to apply the other methods to the Bayesian estimation. Alternatively, the TAP and Bethe approximation methods may be used to approximate the expectation parameters during the iterative procedure of the exact MAP estimation (Eq. 12) because these methods allow us to estimate the expectation parameter from the natural parameters (the forward problem). However, as we found in the estimation of the Fisher information, TAP may occasionally fail and Bethe approximation by BP may not converge. Thus we rather used these methods after the MAP estimation was found by the pseudolikelihood method. We concluded that the sequential Bayes' estimation is more accurately and reliably performed by using the Bethe approximation combining BP and CCCP algorithms. The framework, however, is not limited to these approximation methods, and new methods may be incorporated into the state-space model to further increase the number of neurons that can be analyzed. The large-scale analysis methods would allow us to extrapolate properties of dynamic networks from observed neurons [55, 56].

It should be noted that the current model does not include higher-order interactions to explain the population dynamics. While neural higher-order interactions are ubiquitously observed in vivo [26, 57–59] as well as in vitro [55, 60–62] conditions, it remains to be elucidated how they contribute to characterizing evoked activities. It is an important step to include higher-order interactions in the large-scale time-dependent model. However, the proposed method that includes up to pairwise interactions can be used as a null model for testing activity features involving higher-order interactions. For example, both experimental and modeling studies showed that simultaneous silence of neurons constitutes a major feature of higher-order interactions of stationary neural activities [62, 63]. It remains to be tested, though, if silence probability of all neurons recorded from behaving animals exceed prediction by the pairwise model. Such sparse population activity may be expected when animals are processing natural scenes, compared to artificial stimuli [64].

The limiting factor for the current model on the network size is rather the lack of data than the performance of the approximation methods (Fig. 2). The state-space or other time-resolved methods that include dimension reduction techniques will be important approaches to explain activity of larger populations than analyzed here. The current model serves as a point of reference for fitting performance of these methods. The model goodness-of-fit will be further improved by including covariate signals such as past history of population activities in the model (see [28, 65, 66] for attempts to include these factors in the current model). While applicability to the larger number of neurons and fitting performance of the model can be improved further, the currently proposed method already allows researchers to start testing hypotheses of network responses under distinct task conditions or brain states. These observations will serve to construct biophysical models of neural networks by constraining them, therefore revealing their coding principles.

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Supporting Information

S1 TAP approximation The *Thouless-Anderson-Palmer* (TAP) approach is based on mean-field theory and was first suggested by [43]. There are several ways to derive this approximation [45]. Here we follow the lines of [46, 47] using the *Pleška expansion* [67]. The major difference in our calculation is that $x \in \{0, 1\}$ instead of $\{-1, 1\}$.

The Kullback-Leibler divergence between two probability mass functions is given by:

$$D_{\text{KL}}(q||p) = \sum_{\mathbf{x}} q(\mathbf{x}) \log \frac{q(\mathbf{x})}{p(\mathbf{x})}, \quad (35)$$

For the exponential family distributions $q(\mathbf{x}) = \exp(\boldsymbol{\theta}'_q \mathbf{F}(\mathbf{x}) - \psi)$ and $p(\mathbf{x}) = \exp(\boldsymbol{\theta}'_p \mathbf{F}(\mathbf{x}) - \psi)$, it is written as [68]

$$D_{\text{KL}}(q||p) = \phi(q) - \boldsymbol{\theta}'_p \boldsymbol{\eta}_q + \psi(p), \quad (36)$$

where $\phi(q)$ is the negative entropy of $q(\mathbf{x})$ and $\boldsymbol{\eta}_q = \langle \mathbf{F}(\mathbf{x}) \rangle_q$. $\boldsymbol{\eta}_q$ are alternative coordinates that specify $q(\mathbf{x})$. Here our goal is to find $p(\mathbf{x})$ that minimizes the Kullback-Leibler divergence given $q(\mathbf{x})$. This is equivalent to maximizing $\boldsymbol{\theta}'_p \boldsymbol{\eta}_q - \psi(p)$. If $q(\mathbf{x})$ is an empirical distribution, this is equivalent to the maximum likelihood estimation of $\boldsymbol{\theta}$. Below we identify $\boldsymbol{\theta}_q$ that minimizes the KL divergence given $q(\mathbf{x})$ and hence we will drop the subscripts. Then the negative entropy of $p(\mathbf{x})$ is written as

$$\phi(\boldsymbol{\eta}) = \boldsymbol{\theta}' \boldsymbol{\eta} - \psi(\boldsymbol{\theta}). \quad (37)$$

Taking the derivative of Eq. 37 w.r.t. parameters $\boldsymbol{\theta}$, the function determines the relation between those:

$$\frac{\partial \psi}{\partial \boldsymbol{\theta}} = \boldsymbol{\eta}. \quad (38)$$

Eqs. 37 and 38 are the *Legendre transform*: a translation of a functional relation from $\psi(\boldsymbol{\theta})$ to $\phi(\boldsymbol{\eta})$.

For our model $p(\mathbf{x})$ we now introduce a single scalar α into the distribution which controls the strength of interactions

$$p(\mathbf{x}) = \exp \left(\sum_i \theta_i x_i + \frac{\alpha}{2} \sum_{i \neq j} \theta_{ij} x_i x_j - \psi \right). \quad (39)$$

The model becomes an independent model when $\alpha = 0$. Here the log partition function is a function of $\{\theta_i\}$ and $\{\alpha \theta_{ij}\}$. We now change the variables $\{\theta_i\}$ to $\{\eta_i\}$ by the Legendre transformation of the log partition function to obtain a new free energy:

$$\tilde{\phi}(\{\eta_i\}, \{\alpha \theta_{ij}\}) = \sum_i \theta_i \eta_i - \psi(\{\theta_i\}, \{\alpha \theta_{ij}\}). \quad (40)$$

The function $\tilde{\phi}$ is a function of η_i , θ_{ij} , and α . By assuming weak pairwise interactions because of small α , we approximate $\tilde{\phi}$ by expanding it around the independent model:

$$\tilde{\phi}(\alpha) = \tilde{\phi}|_{\alpha=0} + \left. \frac{\partial \tilde{\phi}}{\partial \alpha} \right|_{\alpha=0} \alpha + \frac{1}{2} \left. \frac{\partial^2 \tilde{\phi}}{\partial \alpha^2} \right|_{\alpha=0} \alpha^2 + \dots \quad (41)$$

The TAP approximation is obtained using expansions up to α^2 . By setting $\alpha = 1$, the approximated free energy $\tilde{\phi}$ is obtained as

$$\begin{aligned} \tilde{\phi}(1) \approx & \sum_{i=1}^N (\eta_i \log \eta_i + (1 - \eta_i) \log(1 - \eta_i)) - \frac{1}{2} \sum_{j \neq i} \theta_{ij} \eta_i \eta_j \\ & - \frac{1}{8} \sum_{j \neq i} \theta_{ij}^2 (\eta_i - \eta_i^2)(\eta_j - \eta_j^2). \end{aligned} \quad (42)$$

This approach is called the Plefka expansion method [67]. The first term is the negative entropy of the independent model whereas the second and third terms are obtained by computing derivatives of the negative entropy w.r.t. α . Derivation of these two terms are given as Eqs. 47 and 49 in the end of this section.

By taking the derivative w.r.t. η_i in Eq. 42, we obtain a system of self-consistent equations

$$\theta_i = \log \left(\frac{\eta_i}{1 - \eta_i} \right) - \sum_{j \neq i} \theta_{ij} \eta_j - \frac{1}{2} \sum_{j \neq i} \theta_{ij}^2 \left(\frac{1}{2} - \eta_i \right) (\eta_j - \eta_j^2). \quad (43)$$

Taking the derivative of Eq. 43 w.r.t. η_j , we obtain the (i, j) element of the inverse Fisher information matrix (for θ_i s):

$$[\mathbf{G}^{-1}]_{ij} = \left(\frac{1}{\eta_i(1-\eta_i)} \right) \delta_{ij} - \theta_{ij} - \theta_{ij}^2 \left(\frac{1}{2} - \eta_i \right) \left(\frac{1}{2} - \eta_j \right). \quad (44)$$

Here δ_{ij} is the Kronecker delta function, where it is 1 if $i = j$ and 0 otherwise. We also let $\theta_{ii} = 0$ for $i = 1, \dots, N$. Using these formulas we can solve the *forward problem*, i.e. given θ obtain an approximation for η . First Eq. 43 is solved numerically to get $\{\eta_i\}$. Then we obtain the upper left part of the inverse Fisher information matrix by Eq. 44 and invert it. By Eq. 15 we see that η_{ij} s are given by

$$\eta_{ij} = [\mathbf{G}]_{ij} + \eta_i \eta_j. \quad (45)$$

Finally, the inverse Legendre transformation yields

$$\begin{aligned} \psi &\approx \sum_i \theta_i \eta_i - \tilde{\phi}(1) \\ &= \sum_i \theta_i \eta_i - \sum_{i=1}^N (\eta_i \log \eta_i + (1 - \eta_i) \log(1 - \eta_i)) + \frac{1}{2} \sum_{j \neq i} \theta_{ij} \eta_i \eta_j \\ &\quad + \frac{1}{8} \sum_{j \neq i} \theta_{ij}^2 (\eta_i - \eta_i^2)(\eta_j - \eta_j^2). \end{aligned} \quad (46)$$

Using $\{\eta_i\}$, we obtain the TAP approximation of the log partition function.

Below we compute derivatives of the negative entropy function. Let the hamiltonian of the system be $H = H^{ext} + \alpha H^{int}$, where $H^{ext} = -\sum_i \theta_i x_i$ and $H^{int} = -\frac{1}{2} \sum_{i \neq j} \theta_{ij} x_i x_j$. We reiterate that $\tilde{\phi}$ is a function of mixture coordinates $(\{\eta_i\}, \{\alpha \theta_{ij}\})$ whereas $\{\theta_i\}$ and H^{ext} are dependent on these parameters.

The first derivative is given as

$$\begin{aligned} \frac{\partial \tilde{\phi}}{\partial \alpha} &= \sum_{i=1}^N \frac{\partial \theta_i}{\partial \alpha} \eta_i - \frac{\partial}{\partial \alpha} \log \sum_{\mathbf{x}} \exp(-H) \\ &= \sum_{i=1}^N \frac{\partial \theta_i}{\partial \alpha} \eta_i + \frac{1}{\sum_{\mathbf{x}} \exp(-H)} \sum_{\mathbf{x}} \exp(-H) \left[H^{int} + \frac{\partial H^{ext}}{\partial \alpha} \right] \\ &= \sum_{i=1}^N \frac{\partial \theta_i}{\partial \alpha} \eta_i + \sum_{\mathbf{x}} \exp(-H - \psi) \left[H^{int} + \frac{\partial H^{ext}}{\partial \alpha} \right] \\ &= \sum_{i=1}^N \frac{\partial \theta_i}{\partial \alpha} \eta_i + \langle H^{int} \rangle_{\alpha} + \left\langle \frac{\partial H^{ext}}{\partial \alpha} \right\rangle_{\alpha} \\ &= \langle H^{int} \rangle_{\alpha}, \end{aligned} \quad (47)$$

where $H = H^{ext} + \alpha H^{int}$ and $\langle \cdot \rangle_{\alpha}$ is the expectation w.r.t. Eq. 39 which depends on α . Substituting $\alpha = 0$ yields

$$\left. \frac{\partial \tilde{\phi}}{\partial \alpha} \right|_{\alpha=0} = \left\langle -\frac{1}{2} \sum_{j \neq i} \theta_{ij} x_i x_j \right\rangle_{\alpha=0} = -\frac{1}{2} \sum_{j \neq i} \theta_{ij} \eta_i \eta_j. \quad (48)$$

The second derivative is given as

$$\begin{aligned}
\frac{\partial^2 \tilde{\phi}}{\partial \alpha^2} &= \frac{\partial \tilde{\phi}}{\partial \alpha} \langle H^{int} \rangle_\alpha = \frac{\partial}{\partial \alpha} \sum_{\mathbf{x}} \exp(-H - \psi) H^{int} \\
&= \sum_{\mathbf{x}} \left[\exp(-H - \psi) \left(-\frac{\partial}{\partial \alpha} H - \frac{\partial}{\partial \alpha} \psi \right) H^{int} \right] \\
&= \left\langle \left(-\frac{\partial}{\partial \alpha} H - \left\langle -\frac{\partial H}{\partial \alpha} \right\rangle_\alpha \right) H^{int} \right\rangle_\alpha \\
&= \left\langle \left(-\frac{\partial}{\partial \alpha} H^{ext} - H^{int} - \left\langle -\frac{\partial}{\partial \alpha} H^{ext} - H^{int} \right\rangle_\alpha \right) H^{int} \right\rangle_\alpha \\
&= \left\langle \left(\sum_i \frac{\partial \theta_i}{\partial \alpha} x_i - H^{int} - \left\langle \sum_i \frac{\partial \theta_i}{\partial \alpha} x_i \right\rangle_\alpha + \langle H^{int} \rangle_\alpha \right) H^{int} \right\rangle_\alpha \\
&= \left\langle \left(\sum_i \frac{\partial \theta_i}{\partial \alpha} (x_i - \eta_i) - H^{int} + \langle H^{int} \rangle_\alpha \right) H^{int} \right\rangle_\alpha.
\end{aligned} \tag{49}$$

Substituting $\alpha = 0$ yields

$$\begin{aligned}
\left. \frac{\partial^2 \tilde{\phi}}{\partial \alpha^2} \right|_{\alpha=0} &= \left\langle \left(\sum_i \frac{\partial \theta_i}{\partial \alpha} (x_i - \eta_i) - H^{int} + \langle H^{int} \rangle_{\alpha=0} \right) H^{int} \right\rangle_{\alpha=0} \\
&= \left\langle \left(-\sum_{j \neq i} \theta_{ij} \eta_j (x_i - \eta_i) + \frac{1}{2} \sum_{j \neq i} \theta_{ij} x_i x_j - \frac{1}{2} \sum_{j \neq i} \theta_{ij} \eta_i \eta_j \right) H^{int} \right\rangle_{\alpha=0} \\
&= -\frac{1}{2} \left\langle \left(-\sum_{j \neq i} \theta_{ij} \eta_j x_i + \frac{1}{2} \sum_{j \neq i} \theta_{ij} \eta_j \eta_i + \frac{1}{2} \sum_{j \neq i} \theta_{ij} x_i x_j \right) \sum_{j \neq i} \theta_{ij} x_i x_j \right\rangle_{\alpha=0} \\
&= -\frac{1}{4} \sum_{j \neq i} \theta_{ij}^2 (\eta_i - \eta_i^2)(\eta_j - \eta_j^2).
\end{aligned} \tag{50}$$

For the last equality we made use of:

$$\begin{aligned}
\left. \frac{\partial \theta_k}{\partial \alpha} \right|_{\alpha=0} &= \left. \frac{\partial^2 \tilde{\phi}}{\partial \alpha \partial \eta_k} \right|_{\alpha=0} = \left. \frac{\partial^2 \tilde{\phi}}{\partial \eta_k \partial \alpha} \right|_{\alpha=0} \\
&= \frac{\partial}{\partial \eta_k} \langle H^{int} \rangle_{\alpha=0} = -\frac{1}{2} \frac{\partial}{\partial \eta_k} \sum_{j \neq i} \theta_{ij} \eta_i \eta_j = -\sum_{j \neq k} \theta_{kj} \eta_j.
\end{aligned} \tag{51}$$

S2 Bethe approximation The Bethe approximation, BP and CCCP are well explained by [44, 48, 49]. However, for the sake of consistency the methods are described here once more. First the Bethe approximation in general will be discussed and subsequently the two algorithms to find its solution.

The Bethe approximation is a variational approach. One assumes that the joint distribution of the Markov network can be written in terms of its individual and pairwise marginals

$$q(\mathbf{x}) = \frac{\prod_{i,j > i} q(x_i, x_j)}{\prod_i q(x_i)^{N_i-1}}, \tag{52}$$

where N_i is the number of neighbors of neuron i . Eq. 52 ignores any cycles in the network and would be exact for a tree. The aim is to find the distribution q that is closest to our actual one p , i.e., the one that minimizes the Kullback-Leibler Divergence

$$D_{KL}(q||p) = \sum_{\mathbf{x}} q(\mathbf{x}) \log \frac{q(\mathbf{x})}{p(\mathbf{x})} = \phi(q) - \boldsymbol{\theta}' \langle \mathbf{F}(\mathbf{x}) \rangle_q + \psi, \tag{53}$$

where $\langle \cdot \rangle_q$ is the expectation over q and $\phi(q)$ its negative entropy. The Bethe approximation of the log partition function is given by

$$\psi \approx \psi_{Bethe} = \psi - D_{KL}(q||p) = -\phi(q) + \theta' \langle \mathbf{F}(\mathbf{x}) \rangle_q. \quad (54)$$

Eq. 54 shows the nature of the approximation error. As long as the class of distribution $q(\mathbf{x})$ contains distributions close to the actual p the error will be small, because the Kullback-Leibler divergence will be small. Furthermore, we see that ψ_{Bethe} will underestimate ψ systematically, because $D_{KL} \geq 0$. Eq. 53 provides an objective function that needs to be minimized w.r.t. q . Realizing that ψ does not depend on $q(\mathbf{x})$, the problem is equivalent to maximizing Eq. 54. Furthermore, $q(x_i)$ and $q(x_i, x_j)$ must fulfill following constraints:

$$q(x_i) = \sum_{x_j} q(x_i, x_j) \text{ for } i = 1, \dots, N, j \neq i \quad (55)$$

Normalization constraints for the marginals are ignored for the moment. The problem can be written as a Lagrangian

$$\mathcal{L}(q) = \psi_{Bethe} + \sum_{i \neq j} \sum_{x_i} \lambda_j(x_i) \left(\sum_{x_j} q(x_i, x_j) - q(x_i) \right). \quad (56)$$

By setting the derivative w.r.t. $q(x_i)$ and $q(x_i, x_j)$ to 0, the marginals can be expressed in terms of the Lagrangian multipliers

$$\begin{aligned} q(x_i, x_j) &\propto \exp(\theta_i x_i + \theta_j x_j + \theta_{ij} x_i x_j + \lambda_j(x_i) + \lambda_i(x_j)), \\ q(x_i) &\propto \exp\left(\theta_i x_i + \frac{\sum_{i \neq j} \lambda_j(x_i)}{N_i - 1}\right). \end{aligned} \quad (57)$$

This constitutes the Bethe approximation and it remains to find the marginals $q(x_i)$ and $q(x_i, x_j)$. In the following subsections two procedures are described, that diverge from this point.

Belief propagation The BP starts from Eq. 57, but writes the Lagrangian multipliers in terms of messages as

$$\lambda_j(x_i) = \log \prod_{k \in N(i) \setminus j} m_k(x_i). \quad (58)$$

$N(i) \setminus j$ are the set of neighbors of i without j and $m_k(x_i)$ is the *message* sent from node k to i . Substituting this into Eq. 57 yields

$$\begin{aligned} q(x_i, x_j) &\propto \exp(\theta_i x_i + \theta_j x_j + \theta_{ij} x_i x_j) \prod_{k \in N(i) \setminus j} m_k(x_i) \prod_{k \in N(j) \setminus i} m_k(x_j), \\ q(x_i) &\propto \exp(\theta_i x_i) \prod_{k \in N(i)} m_k(x_i). \end{aligned} \quad (59)$$

By substituting these marginals into Eq. 55 a set of self-consistent equations for the messages can be obtained

$$m_j(x_i) = \sum_{x_j} \exp(\theta_j x_j + \theta_{ij} x_i x_j) \prod_{k \in N(j) \setminus i} m_k(x_j). \quad (60)$$

The BP algorithm initializes the messages and solves Eq. 60 iteratively until the algorithm converges. Having obtained the messages the marginals can be computed by Eq. 59 and they just need to be normalized in the end.

Concave convex procedure While the BP algorithm takes only care of the normalization constraints in the end and hence does sometimes not converge, the CCCP [49] is more strictly about them, which guarantees convergence at the cost of computation time.

The starting point is the Lagrangian function depicted in Eq. 56. The difference is that in [49] instead of maximizing ψ_{Bethe} they minimize the so-called Gibbs free energy which is just $-\psi_{Bethe}$. Furthermore, the normalization constraint

$$\sum_{x_i, x_j} q(x_i, x_j) = 1, \quad (61)$$

is added, resulting in the Lagrangian

$$\mathcal{L}_{CCCP}(q) = -\psi_{Bethe} + \sum_{i \neq j} \sum_{x_i} \lambda_j(x_i) \left(\sum_{x_j} q(x_i, x_j) + q(x_i) \right) + \sum_{i \neq j} \gamma_{ij} \left(\sum_{x_i, x_j} q(x_i, x_j) - 1 \right). \quad (62)$$

The basic principle of the CCCP is to realize that $-\psi_{Bethe}$ can be decomposed into a convex and a concave part

$$\begin{aligned} -\psi_{Bethe} = & \underbrace{\sum_{i \neq j} \sum_{x_i, x_j} q(x_i, x_j) \log \frac{q(x_i, x_j)}{\exp(\theta_i x_i + \theta_j x_j + \theta_{ij} x_i x_j)} + \sum_i \sum_{x_i} q(x_i) \log \frac{q(x_i)}{\exp(\theta_i x_i)}}_{F_{convex}} \\ & - \underbrace{\sum_i N_i \sum_{x_i} q(x_i) \log \frac{q(x_i)}{\exp(\theta_i x_i)}}_{F_{concave}}. \end{aligned} \quad (63)$$

Calculating the derivative w.r.t. the marginals yields the following iterative update rule for q

$$\begin{aligned} \frac{\partial}{\partial q(x_i, x_j)} F_{convex}(q^{t+1}) &= - \frac{\partial}{\partial q(x_i, x_j)} F_{concave}(q^t) - \lambda_i(x_j) - \lambda_j(x_i) - \gamma_{ij}, \\ \frac{\partial}{\partial q(x_i)} F_{convex}(q^{t+1}) &= - \frac{\partial}{\partial q(x_i)} F_{concave}(q^t) + \sum_k \lambda_k(x_i). \end{aligned} \quad (64)$$

By updating the marginals with Eq. 64 $-\psi_{Bethe}$ is monotonically decreased (see Theorem 2 in [49]). Writing the update explicitly for the marginals we get

$$\begin{aligned} q^{t+1}(x_i, x_j) &= \exp(\theta_i x_i + \theta_j x_j + \theta_{ij} x_i x_j - \lambda_i(x_j) - \lambda_j(x_i) - \gamma_{ij}), \\ q^{t+1}(x_i) &= \left(\frac{q^t(x_i)}{\exp(\theta_i x_i)} \right)^{N_i} \exp \left(\theta_i x_i + N_i + \sum_j \lambda_j(x_i) - 1 \right). \end{aligned} \quad (65)$$

Note, that here t is an integer describing the iterations of the algorithm and not the time-dependence of the model. Assume we have a set of Lagrangian multipliers such that the constraints in Eq. 55 and 61 are satisfied. Then $-\psi_{Bethe}$ can be decreased by updating the marginals with Eq. 65. However, by doing so the constraints will be violated and one has to update the Lagrangian multipliers. By substituting Eq. 65 into the constraints (Eq. 55 and 61) one gets self-consistent equations for the multipliers that write as

$$\begin{aligned} \exp(\gamma_{ij}) &= \sum_{x_i, x_j} \exp(\theta_i x_i + \theta_j x_j + \theta_{ij} x_i x_j - \lambda_j(x_i) - \lambda_i(x_j) - 1), \\ \exp(2\lambda_j(x_i)) &= \frac{\sum_{x_j} \exp(\theta_j x_j + \theta_{ij} x_i x_j - \lambda_i(x_j) - \gamma_{ij})}{\left(\frac{q^t(x_i)}{\exp(\theta_i x_i)} \right)^{N_i} \exp \left(N_i + \sum_{k \neq \{i, j\}} \lambda_k(x_i) \right)}. \end{aligned} \quad (66)$$

The multipliers are updated sequentially until the constraints for the marginals are again satisfied.

The CCCP always updates first the marginals. For each update the Lagrangian multipliers have to be updated until the constraints are fulfilled again. This alternating procedure is done until the Bethe free energy converges.

S3 Data generation Here we explain how the underlying model parameters for Fig. 1-4 are generated, and the artificial spike data is sampled from the model. We discuss the model parameters used to generate the subpopulation activity. The underlying time-varying parameters $\theta_{1:T}$ are sampled as Gaussian processes of $T = 500$ time bins, for $i, j = 1, \dots, N$:

$$\theta_i^{1:T} \sim \mathcal{GP}(\boldsymbol{\mu}, \mathbf{K}), \quad (67)$$

$$\theta_{ij}^{1:T} \sim \mathcal{GP}(\mathbf{0}, \mathbf{K}), \quad (68)$$

where $\boldsymbol{\mu}$ is a mean vector of size T , and $\mathbf{K}$ is the $T \times T$ covariance matrix. For $\theta_{ij}^{1:T}$, the mean is fixed at zero. For $\theta_i^{1:T}$, the mean vector $\boldsymbol{\mu} = (\mu_1, \dots, \mu_T)$ is modulated using an inverse Gaussian function as

$$\mu_t = \begin{cases} -2 & \text{for } t < 100 \\ -2 + \frac{\lambda}{2\pi g(t)^3} \exp\left(-\frac{\lambda}{2f(t)}(g(t) - 1)^2\right) & \text{for } t \geq 100, \end{cases} \quad (69)$$

where $g(t) = 3(t - 100)/400$ and $\lambda = 3$. To produce smooth processes, the covariance matrix $\mathbf{K}$ dictating the smoothness for both θ_i and θ_{ij} is chosen as

$$[\mathbf{K}]_{t,t'} = \frac{1}{\sigma_1} \exp\left(-\frac{|t - t'|^2}{2\sigma_2^2}\right), \quad (70)$$

where $\sigma_1 = 12$ and $\sigma_2 = 50$. While the processes of the first order natural parameters $\{\theta_i^{1:T}\}$ have time-varying mean at different time points, it should be noted that the sampled interactions $\{\theta_{ij}^{1:T}\}$ also smoothly change over time.

Using sampled underlying parameters for the observation model, 200 trials of spiking data are sampled from the resulting probability mass function by the Gibbs sampling method. Given $\theta_{1:T}$ each time bin can be sampled independently. The first 100 samples are omitted and then every 10th sample is chosen such that samples are approximately independent.

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